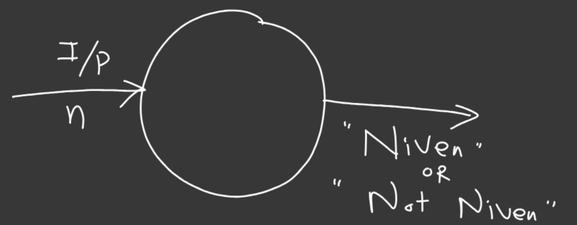
Niven Number.

A number is said to be a "Niven" Number if it is divisible by sum of its digits.

e.g:- Let n = 18

sum of digits = 1+8 = 9

e.g:- 24, 27 . . . 18 % 9 = 0 ⟶ So 18 is a Niven Number.



optional optional

```
for (Initializtn ; Condn ; Incr/Dcr)
```

```
for( ; n!=0; n=n/10)
{
    r = n % 10;
    sum = sum+r;
}
```

```
int n = sc.nextInt();
int sum=0, r;
int num=n;
while (n!=0)
{
    r = n % 10;
    sum = sum+r;
    n = n/10;
}
```

sum of digits

120 → num, n

0 → 12 → 120 → 0

r: 0 → 0 → 2 → 1

sum: 0 → 2 → 3

n: 120 → 12 → 1 → 0

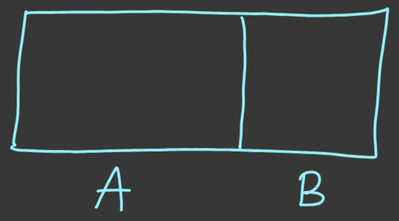
n % 10: 120 % 10 → 12 % 10 → 1 % 10

n/10: 120/10 → 12/10 → 1/10

```
if (num % sum == 0)
    S.o.pln ("Niven Number");
else
    S.o.pln ("Not Niven Number");
```

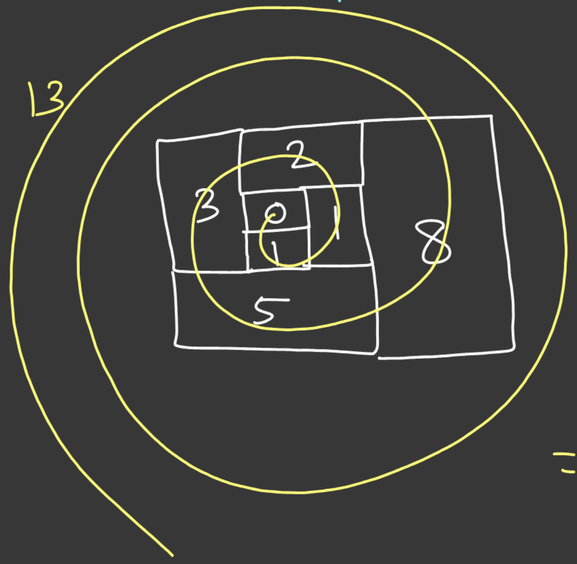
num % sum: 120 % 3

Golden Ratio = 1.61



$$\frac{A+B}{A} = 1.61$$





$5 = 2+3$

$(A+B)\ 5/3 = 1.6$

$8/5 = 1.6$

$13/8 =$

Fibonacci Number

New Fibonacci Number

= Addⁿ of Previous two fibonacci Number

0 1 1 2 3 5 8 13 21 34 55 ....

F_1 F_2 F_3

$F_n = F_{n-1} + F_{n-2}$

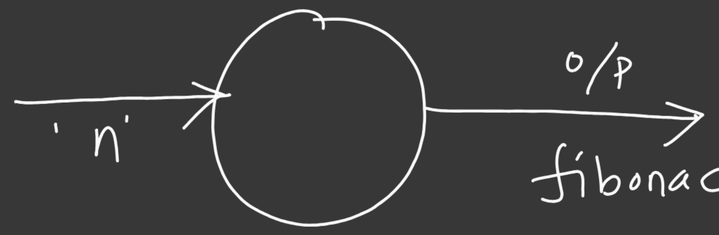
$F_3 = F_2 + F_1$

given. $1 \quad 1 + 0$

$F_4 = F_3 + F_2$

$2 = 1 + 1$

Q:- To generate 1st 'n' term of the fibonacci series.

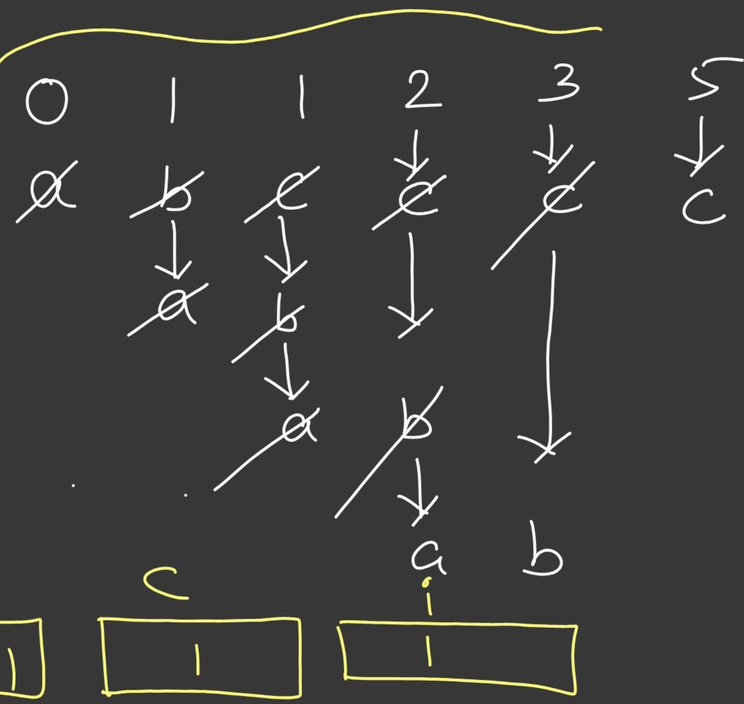


Let n=8 ⟶ 0 1 1 2 3 5 8 13

```
int n=sc.nextInt();
int a=0, b=1, c;
for(int i=1; i<n; i++)
{
    c=a+b;
    a=b;
    b=c;
}
System.out.println(a);
```

Let a=0
b=1

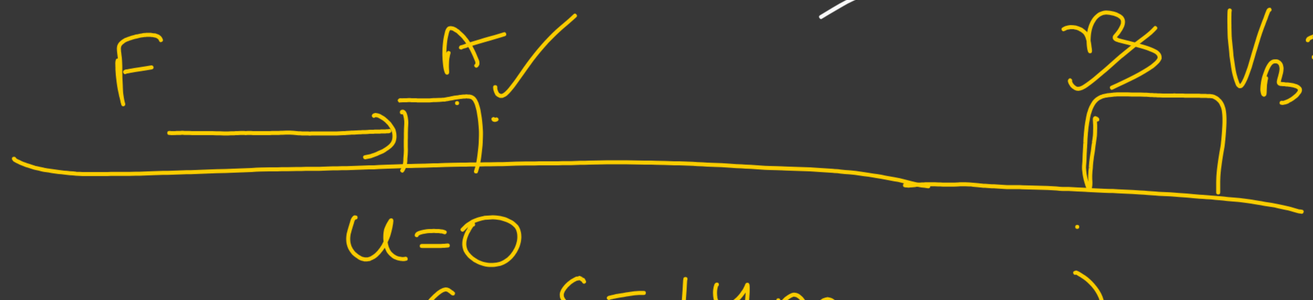
c=a+b
a=b
b=c





1

Ex 6.47 $F(x) = 18 - 0.530x$ is applied on 6 kg body initially at rest on a horizontal frictionless surface. What is its speed when it travels 14 m.



$$W = \int \vec{F} \cdot d\vec{x} = \int_0^{14} (18 - 0.530x)\,dx = \left[18x - 0.530\frac{x^2}{2}\right]_0^{14}$$

$$WET = \left[18x - 0.530\frac{x^2}{2}\right]_0^{14}$$

$$\Rightarrow W = 200.06\,J = \Delta KE = K_f - K_i = \frac{1}{2}mV_B^2 - \frac{1}{2}mV_A^2$$

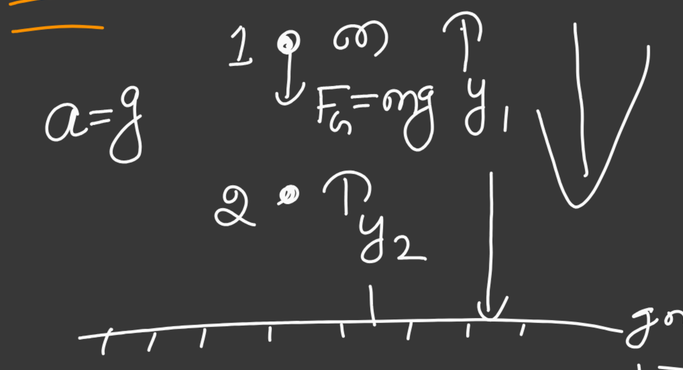
$$\Rightarrow 200.06 = \frac{1}{2} \times 6 \times V_B - \frac{1}{2} \times 6 \times 0^2$$

$$\Rightarrow V_B = 8.017\ m/s$$

Ch-07 (Energy)

PE ((U)) ← Position configuration

Q P.t conservation of ME in Grav. system



$$W_{grav} = \int_{-y_1}^{-y_2} mg\, dy = mg[y]_{-y_1}^{-y_2}$$

$$\Rightarrow W_{grav} = mg[-y_2 - (-y_1)]$$

$$\Rightarrow W_{grav} = mgy_1 - mgy_2$$

$$= U_i - U_f$$

$W_{grav} = +ve$



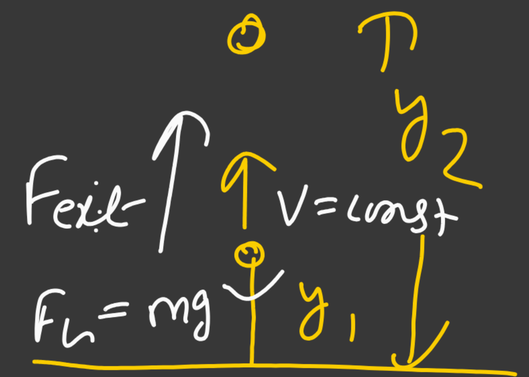
$$W_{grav} = \int dW = \int_{y_2}^{y_1} (mg)\, dy \cos 180$$

$$= \int -mg\, dy = -mg[y]_{y_2}^{y_1}$$

$$= -mg[y_2 - y_1]$$

$$\Rightarrow W_{grav} = mgy_1 - mgy_2$$

$$= -ve$$



$$W_{grav} = mgy_1 - mgy_2$$

$F_{ext} = mg$

$$W_{ext} = -W_{grav} = -\left[mgy_1 - mgy_2\right]$$

$$\Rightarrow W_{ext} = mgy_2 - mgy_1$$

$$= U_{grav,2} - U_{grav,1}$$

$$W_{ext} = \Delta U_{grav}$$

15 → 20%

70%

$ME = PE + KE$

$\Rightarrow E = U_{grav} + K = \text{const}$ under Grav. force only

As per WET, $W_{Total} = \Delta KE$

$= K_f - K_i$

$W_{grav} = \underbrace{mgy_1 - mgy_2}$

$= W_{Total}$

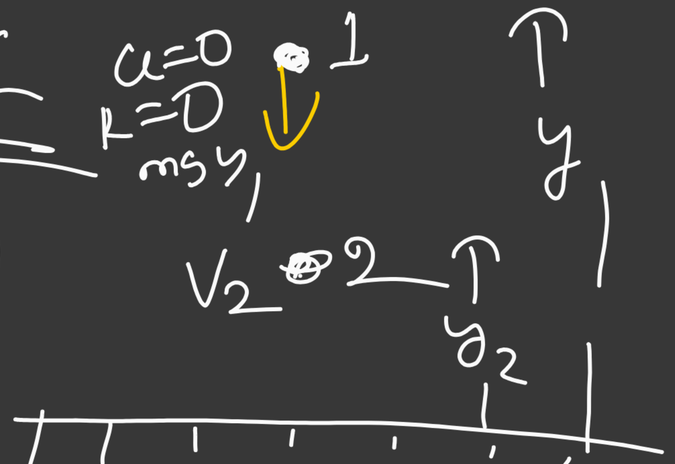
$= K_f - K_i = mgy_1 - mgy_2$

$= U_{grav,1} - U_{grav,2}$

$\Rightarrow \underbrace{K_f + U_{grav,2}} = K_i + U_{grav,1}$

$ME_f = ME_i$

Proof



$0 + mgy_1$

$\frac{1}{2}mv_2^2 + mgy_2$

$= mgy_1$

$ME = \text{const}$

Case (2) Other than Gravitational force, if other forces are present.

as per WET $W_{net} = \Delta KE$

$$\Rightarrow W_{other} + W_{grav} = K_f - K_i$$

$$\Rightarrow W_{other} + (mgy_1 - mgy_2) = K_f - K_i$$

$$\Rightarrow W_{other} + (U_{grav,1} - U_{grav,2}) = K_2 - K_1$$

$$\Rightarrow W_{other} = (U_{grav,2} + K_2) - (U_{grav,1} + K_1)$$

$$\boxed{W_{other} = E_2 - E_1}$$